- Assign the connection name and the IP address and choose the "Used interface" and continue with "Next".



Fig. 15: OC Wizard – Set properties of "Communication partner A"

Normally the port of the PNOZ multi Ethernet modul is the port "9000".

- If the port is changed, please correct this parameter and go on with "Next".

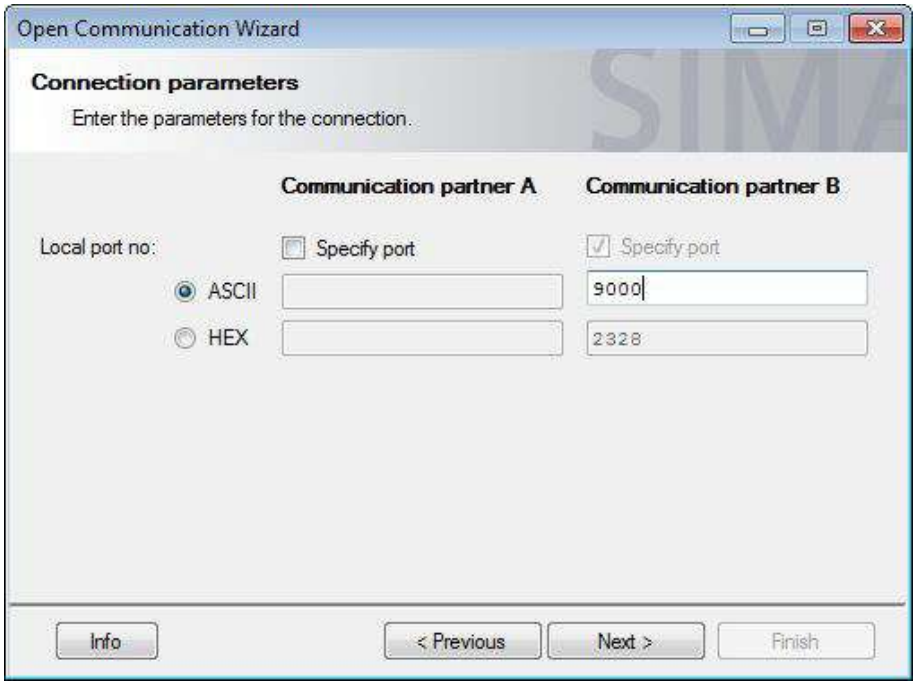


Fig. 16: OC Wizard – Set "Connections parameters" of "Communication partner B"

- Declare any UDT. In this example "UDT 65" and continue with “Next”.

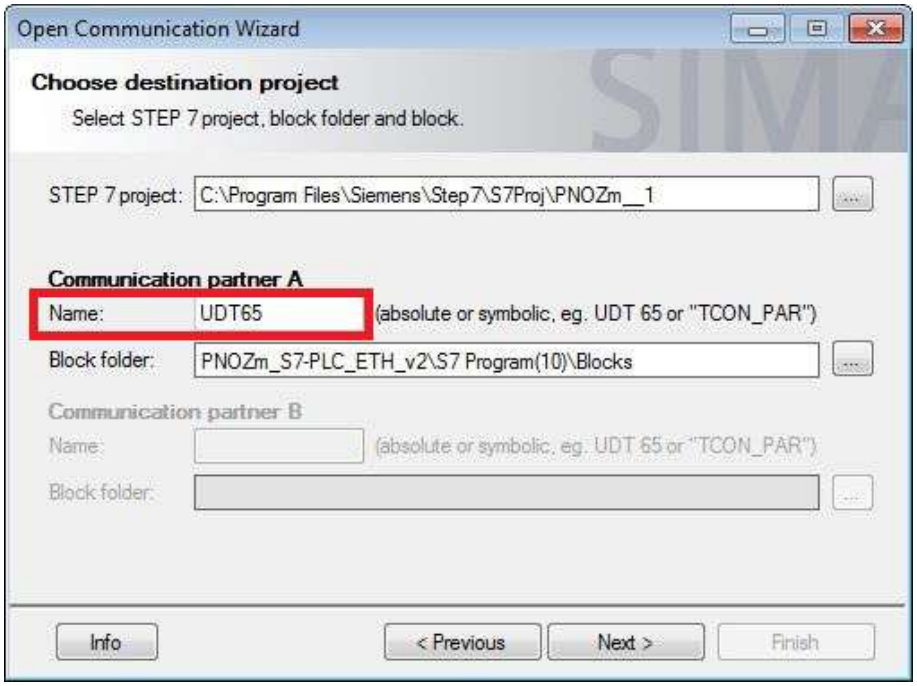


Fig. 17: OC Wizard – Choose destination project, block folder and block “UDT65”

- Control all properties of new connection and continue with "Next".

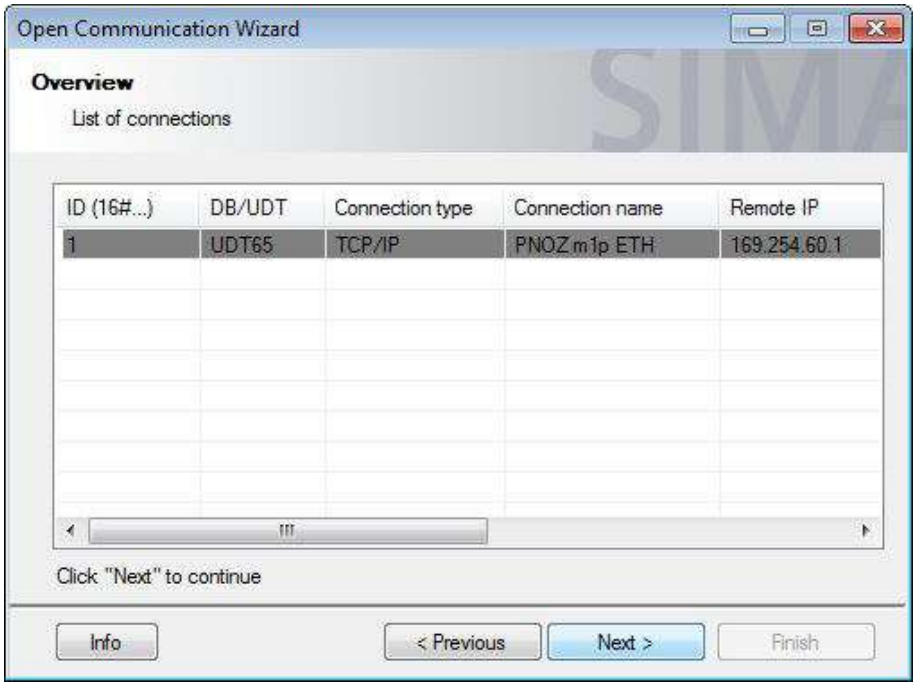


Fig. 18: OC Wizard – Overview with “List of Connections” for “UDT65”

In the S7 project will be set “UDT 65”:

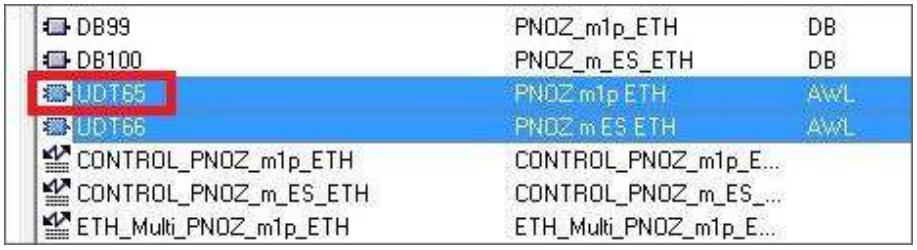


Fig. 19: Simatic Manager – Step7-Project with List of blocks (“UDT65”)

- Confirm with “Finish” and close the “Open Communication Wizard”.

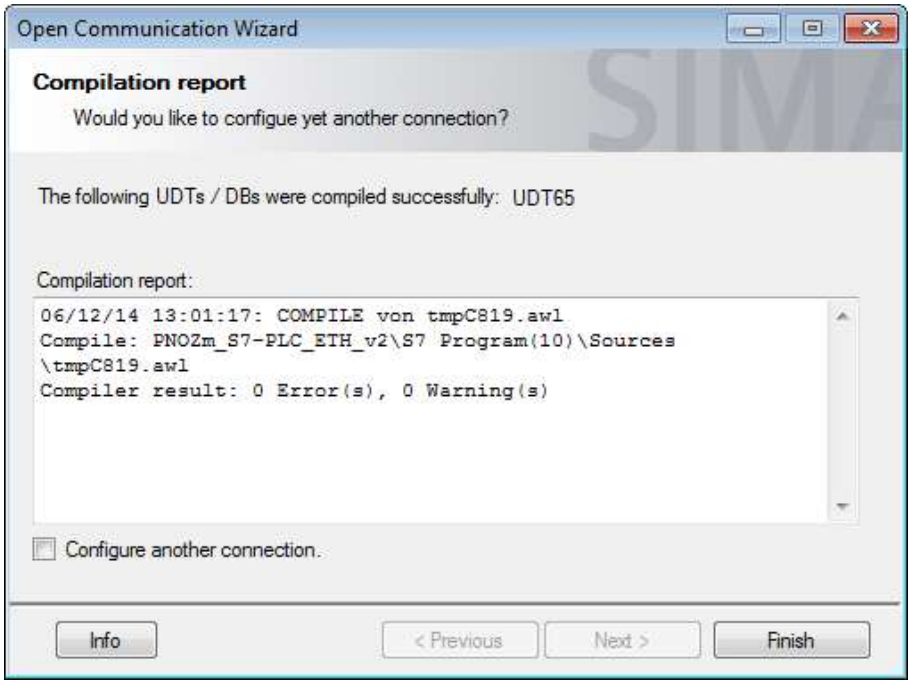


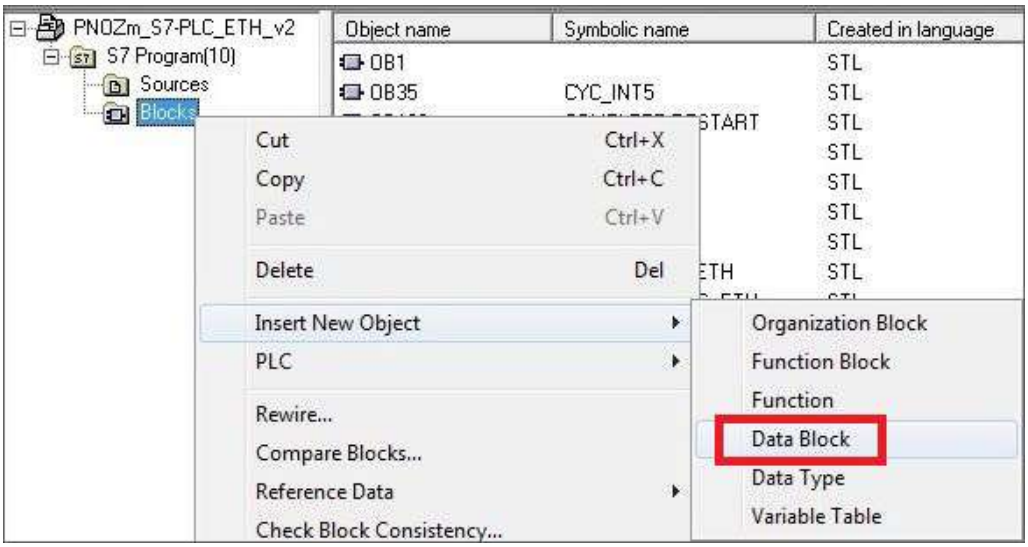


Fig. 20: OC Wizard – Compilation Report and Confirm finished of setup

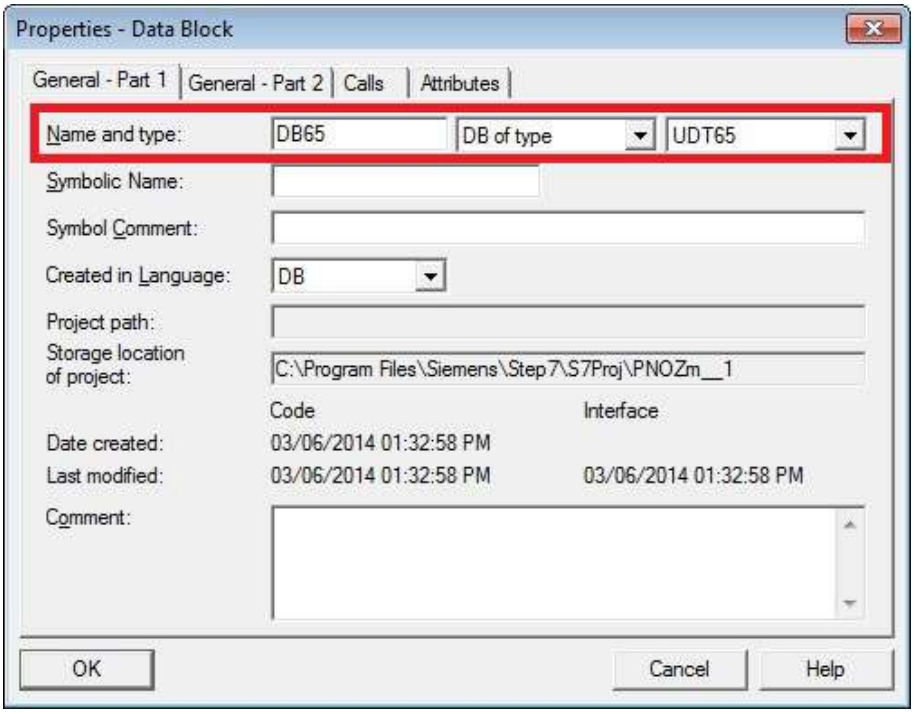
4.4. Complete and Download actual Step7-Project

- Open your actual Step7-Project (here as example “ETH_Multi_S7_ProfiNET_v2”) in the Simatic Manager.
- Now you have to insert a new data block.
  - For this step click with the right mouse button at “Blocks” (German: “Bausteine”), choose "Insert New Object" and click at "Data Block".
  - For properties set name “DB65”, "DB of type" of the data block and “UDT 65”.

Step 1:



Step 2:



Step 3:



Fig. 21: Simatic Manager – Create and set properties of Data Block “DB65” of “UDT65”

- Open Function Block "FC3" for PNOZ m1p ETH, PNOZ mmc1p / “FC4” for PNOZ m ES ETH and change the ID in all network parts.

```
// ==== CONNECT ====================================

      CALL  "TCON" , "InstanzDB-FB65"     FB65 / DB10        -- Connect
       REQ     :=M10.0
       ID      :=W#16#1
       DONE    :=M10.1
       BUSY    :=M10.2
       ERROR   :=M10.3
       STATUS  :=MW11
       CONNECT:=P#DB65.DBX0.0 BYTE 64
```

Fig. 22: AWL-Editor – Change ID in Function Block “FC3”, section “CONNECT”

```
// ==== SEND ====================================

      CALL  "TSEND" , "InstanzDB-FB63"          FB63 / DB11        -- Send Data
       REQ     :=M20.0
       ID      :=W#16#1
       LEN     :=30
       DONE    :=M20.1
       BUSY    :=M20.2
       ERROR   :=M20.3
       STATUS:=MW21
       DATA    :="PNOZ m1p ETH SEND-DATA".DB_VAR    P#DB6.DBX0.0
```

Fig. 23: AWL-Editor – Change ID in Function Block “FC3”, section “SEND”

```
// ==== RECEIVE ====================================

      CALL  "TRCV" , "InstanzDB-FB64"           FB64 / DB13        -- Receive Data
       EN_R    :=M40.0
       ID      :=W#16#1
       LEN     :=30
       NDR     :=M40.1
       BUSY    :=M40.2
       ERROR   :=M40.3
       STATUS  :=MW41
       RCVD_LEN:=MW43
       DATA    :="PNOZ m1p ETH RCV-DATA".DB_VAR    P#DB7.DBX0.0
```

Fig. 24: AWL-Editor – Change ID in Function Block “FC3”, section “RECEIVE”

```
// ==== DISCONNECT ====================================

      CALL  "TDISCON" , "InstanzDB-FB66"    FB66 / DB12        -- Disconnect
       REQ     :=M30.0
       ID      :=W#16#1
       DONE    :=M30.1
       BUSY    :=M30.2
       ERROR   :=M30.3
       STATUS:=MW31
```

Fig. 25: AWL-Editor – Change ID in Function Block “FC3”, section “DISCONNECT”

- Save modified Function Block "FC3" and download complete Step7-Project to Simatic PLC:

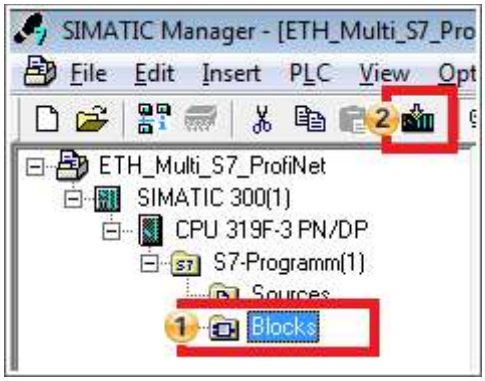


Fig. 26: Simatic Manager – Start complete download of Step7-Project

- For next message box choose “All”:

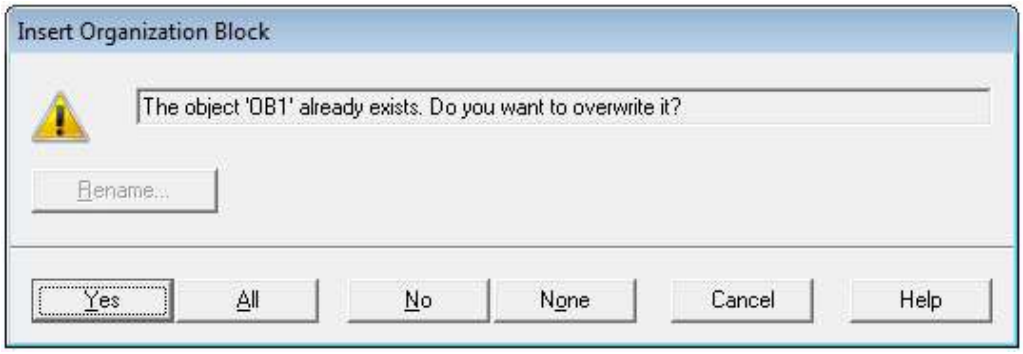


Fig. 27: Simatic Manager – Download, Message box “Insert Organization Block”

- For next four message boxes choose “Yes” or “OK”:

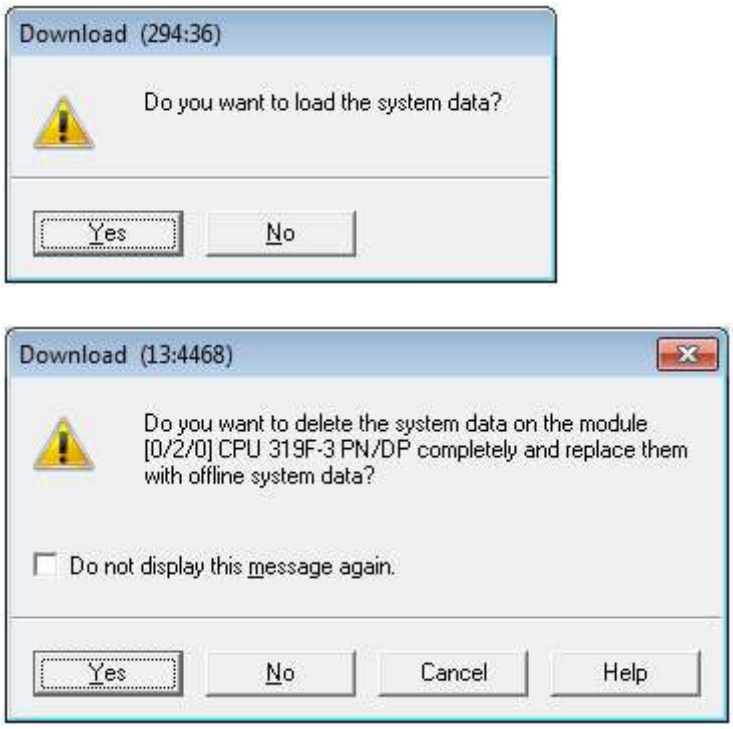


Fig. 28: Simatic Manager – Download, Message boxes “Download”

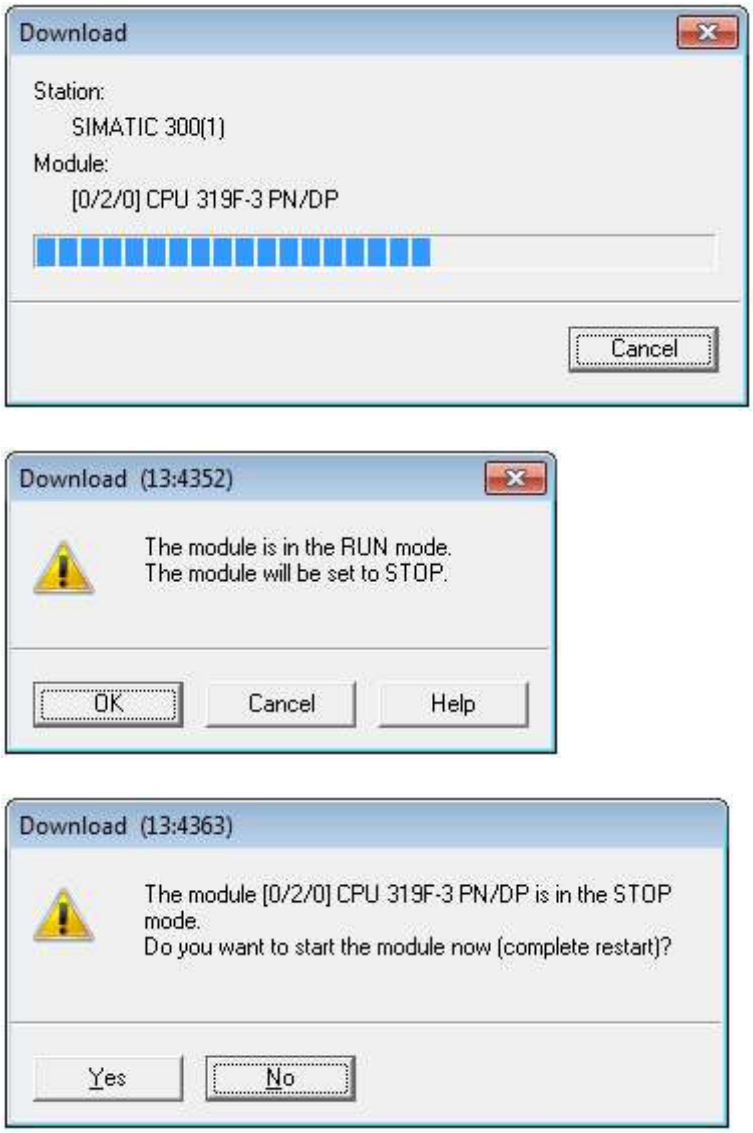


Fig. 29: Simatic Manager – Download, Message boxes "Download"

4.5. Check successful communication via Monitoring

4.5.1. Monitoring on Simatic PLC

- Open both variable tables “CONTROL_PNOZ_m1p_ETH” and “ETH_Multi_PNOZ_m1p_ETH” and Activate for each variable table function “Monitor variable”.

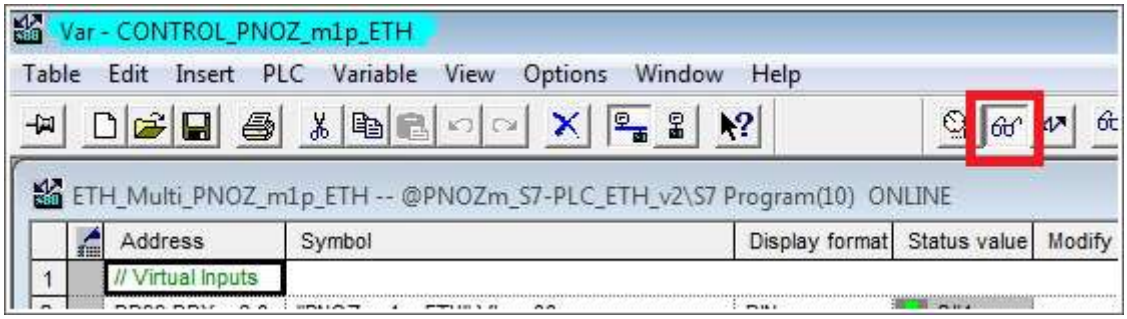


Fig. 30: Var Table – Activate monitoring of variables

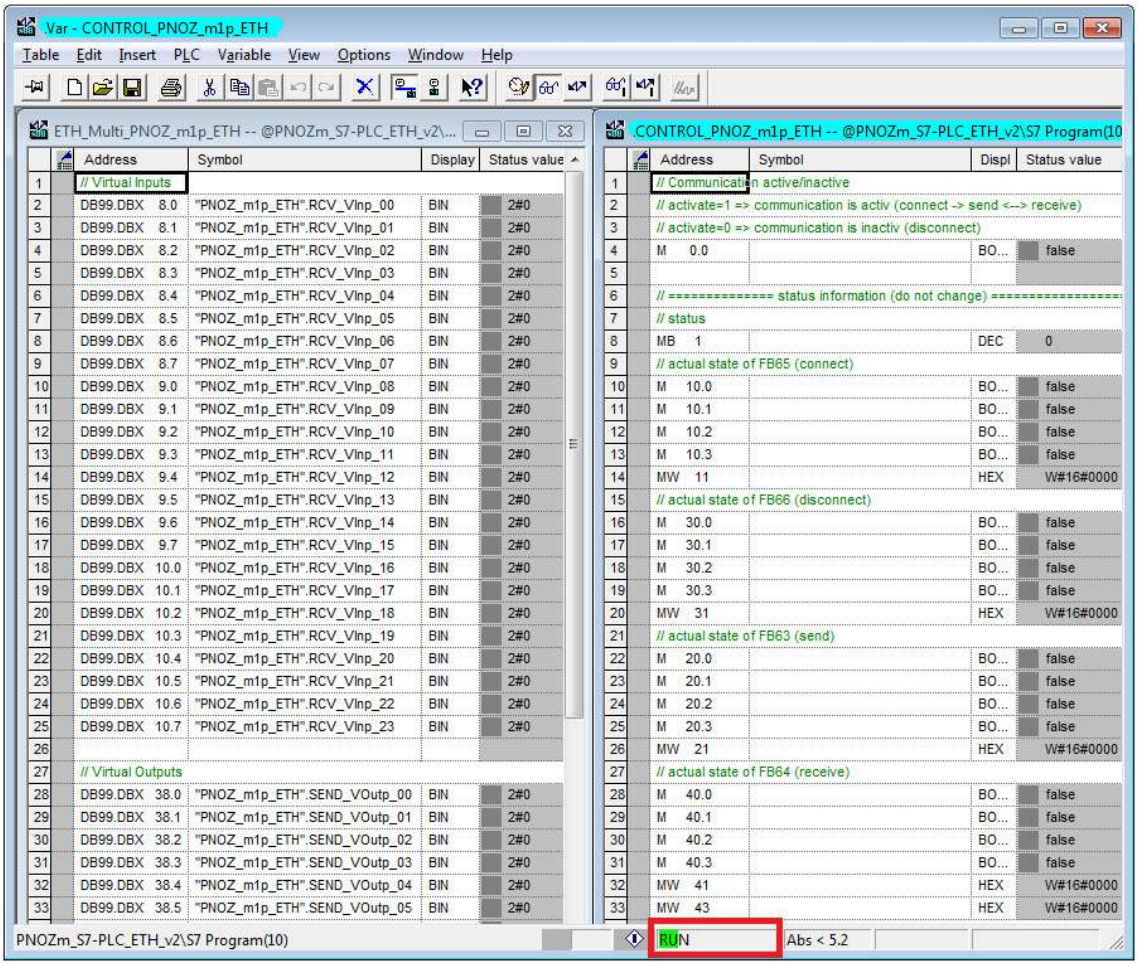


Fig. 31: Var Table – Monitoring of variables is active

- Activate the “CONTROL_PNOZ_m1p_ETH” window, click with your right mouse button at address “M0.0” on Symbol “activate” and set it to ”Modify Address to 1”.

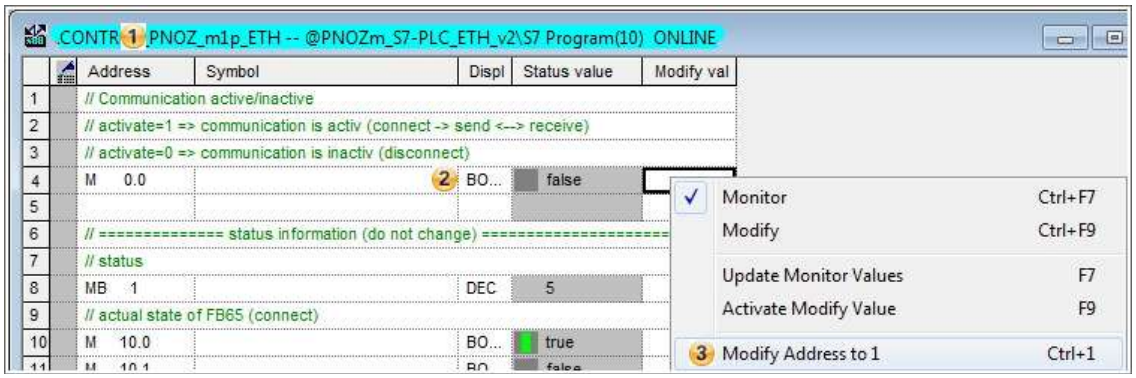


Fig. 32: Var Table – Modify Address “M0.0” to =1

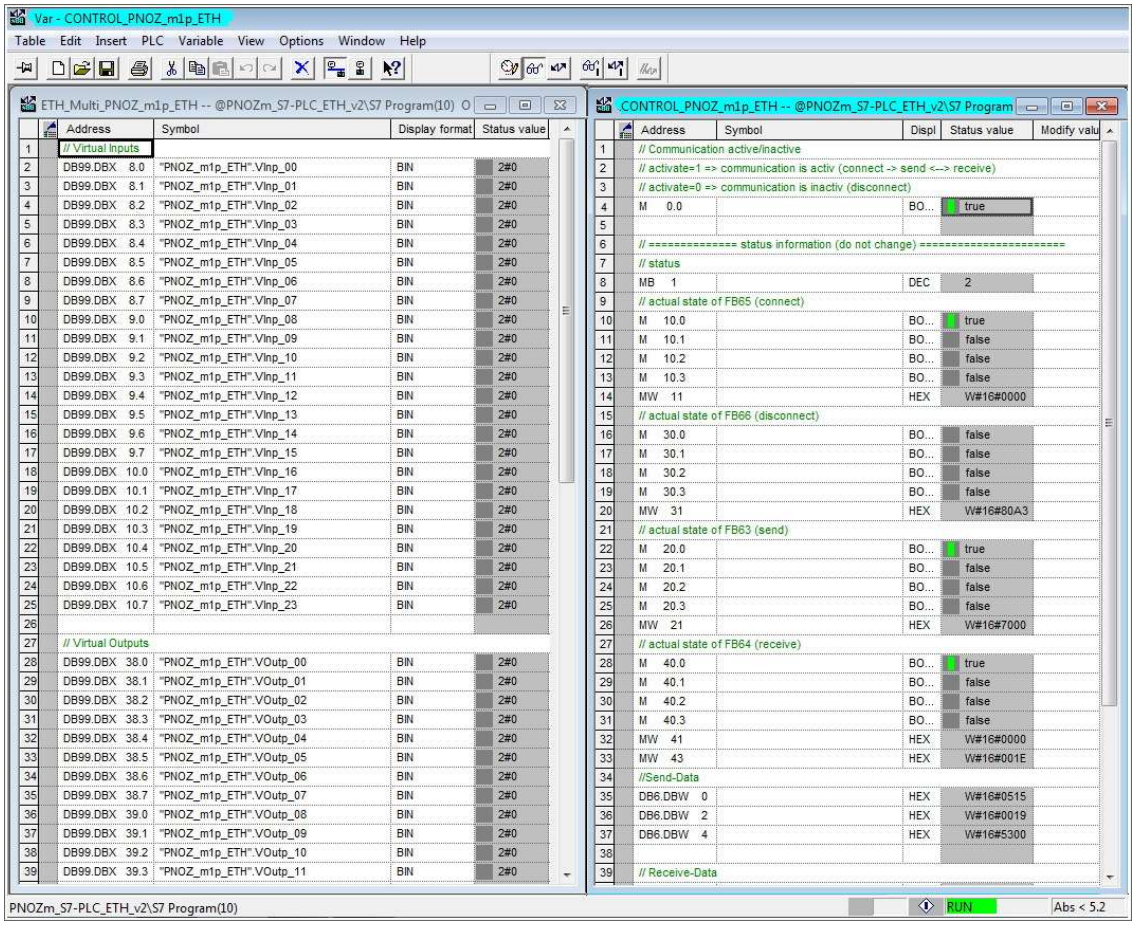


Fig. 33: Var Table – Monitoring of variables is active

- Activate the “ETH_Multi_PNOZ_m1p_ETH” window, select the Virtual Output 00, 08 and 16 and write in the “Modify value” 2#1.

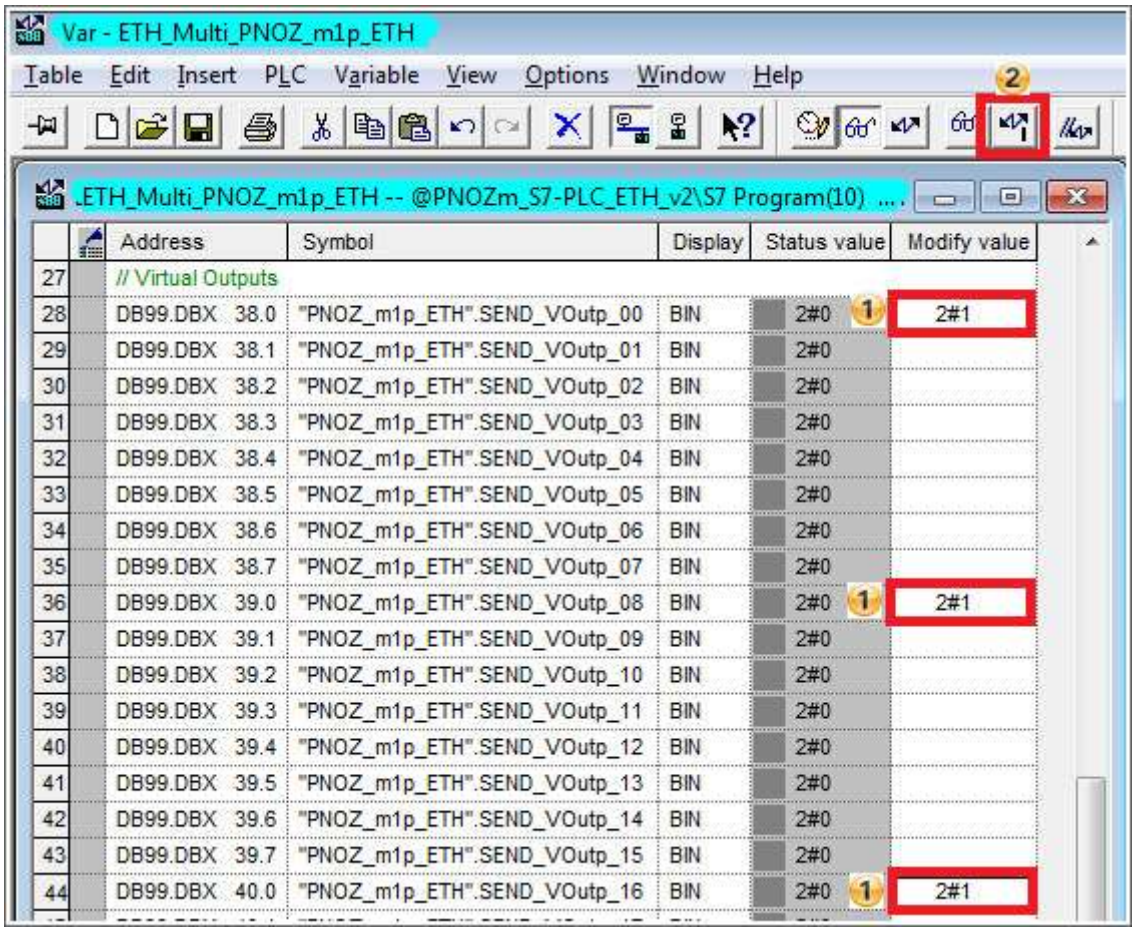


Fig. 34: Var Table – Modify Address Virtual Outputs to =1

The according bits will be set to the signal “=1”.
These tree bits are mirrored in the PNOZ multi application program. When a Virtual Output bit is set to “=1”, the Virtual Input bit is also “=1”.

27		// Virtual Outputs					
28		DB99.DBX 38.0	"PNOZ_m1p_ETH".SEND_VOutp_00	BIN		2#1	2#1
29		DB99.DBX 38.1	"PNOZ_m1p_ETH".SEND_VOutp_01	BIN		2#0	
30		DB99.DBX 38.2	"PNOZ_m1p_ETH".SEND_VOutp_02	BIN		2#0	
31		DB99.DBX 38.3	"PNOZ_m1p_ETH".SEND_VOutp_03	BIN		2#0	
32		DB99.DBX 38.4	"PNOZ_m1p_ETH".SEND_VOutp_04	BIN		2#0	
33		DB99.DBX 38.5	"PNOZ_m1p_ETH".SEND_VOutp_05	BIN		2#0	
34		DB99.DBX 38.6	"PNOZ_m1p_ETH".SEND_VOutp_06	BIN		2#0	
35		DB99.DBX 38.7	"PNOZ_m1p_ETH".SEND_VOutp_07	BIN		2#0	
36		DB99.DBX 39.0	"PNOZ_m1p_ETH".SEND_VOutp_08	BIN		2#1	2#1
37		DB99.DBX 39.1	"PNOZ_m1p_ETH".SEND_VOutp_09	BIN		2#0	
38		DB99.DBX 39.2	"PNOZ_m1p_ETH".SEND_VOutp_10	BIN		2#0	
39		DB99.DBX 39.3	"PNOZ_m1p_ETH".SEND_VOutp_11	BIN		2#0	
40		DB99.DBX 39.4	"PNOZ_m1p_ETH".SEND_VOutp_12	BIN		2#0	
41		DB99.DBX 39.5	"PNOZ_m1p_ETH".SEND_VOutp_13	BIN		2#0	
42		DB99.DBX 39.6	"PNOZ_m1p_ETH".SEND_VOutp_14	BIN		2#0	
43		DB99.DBX 39.7	"PNOZ_m1p_ETH".SEND_VOutp_15	BIN		2#0	
44		DB99.DBX 40.0	"PNOZ_m1p_ETH".SEND_VOutp_16	BIN		2#1	2#1
1		// Virtual Inputs					
2		DB99.DBX 8.0	"PNOZ_m1p_ETH".RCV_VInp_00	BIN		2#1	
3		DB99.DBX 8.1	"PNOZ_m1p_ETH".RCV_VInp_01	BIN		2#0	
4		DB99.DBX 8.2	"PNOZ_m1p_ETH".RCV_VInp_02	BIN		2#0	
5		DB99.DBX 8.3	"PNOZ_m1p_ETH".RCV_VInp_03	BIN		2#0	
6		DB99.DBX 8.4	"PNOZ_m1p_ETH".RCV_VInp_04	BIN		2#0	
7		DB99.DBX 8.5	"PNOZ_m1p_ETH".RCV_VInp_05	BIN		2#0	
8		DB99.DBX 8.6	"PNOZ_m1p_ETH".RCV_VInp_06	BIN		2#0	
9		DB99.DBX 8.7	"PNOZ_m1p_ETH".RCV_VInp_07	BIN		2#0	
10		DB99.DBX 9.0	"PNOZ_m1p_ETH".RCV_VInp_08	BIN		2#1	
11		DB99.DBX 9.1	"PNOZ_m1p_ETH".RCV_VInp_09	BIN		2#0	
12		DB99.DBX 9.2	"PNOZ_m1p_ETH".RCV_VInp_10	BIN		2#0	
13		DB99.DBX 9.3	"PNOZ_m1p_ETH".RCV_VInp_11	BIN		2#0	
14		DB99.DBX 9.4	"PNOZ_m1p_ETH".RCV_VInp_12	BIN		2#0	
15		DB99.DBX 9.5	"PNOZ_m1p_ETH".RCV_VInp_13	BIN		2#0	
16		DB99.DBX 9.6	"PNOZ_m1p_ETH".RCV_VInp_14	BIN		2#0	
17		DB99.DBX 9.7	"PNOZ_m1p_ETH".RCV_VInp_15	BIN		2#0	
18		DB99.DBX 10.0	"PNOZ_m1p_ETH".RCV_VInp_16	BIN		2#1	

Fig. 35: Var Table – Virtual In/Outputs

4.5.2. Monitoring on PNOZmulti base unit

- Open the provided PNOZmulti-Project (“PNOZmulti_S7-PLC_ETH-TCP-IP_v2”) with Software PNOZmulti Configurator; the password for “Level 1” is number “1”.

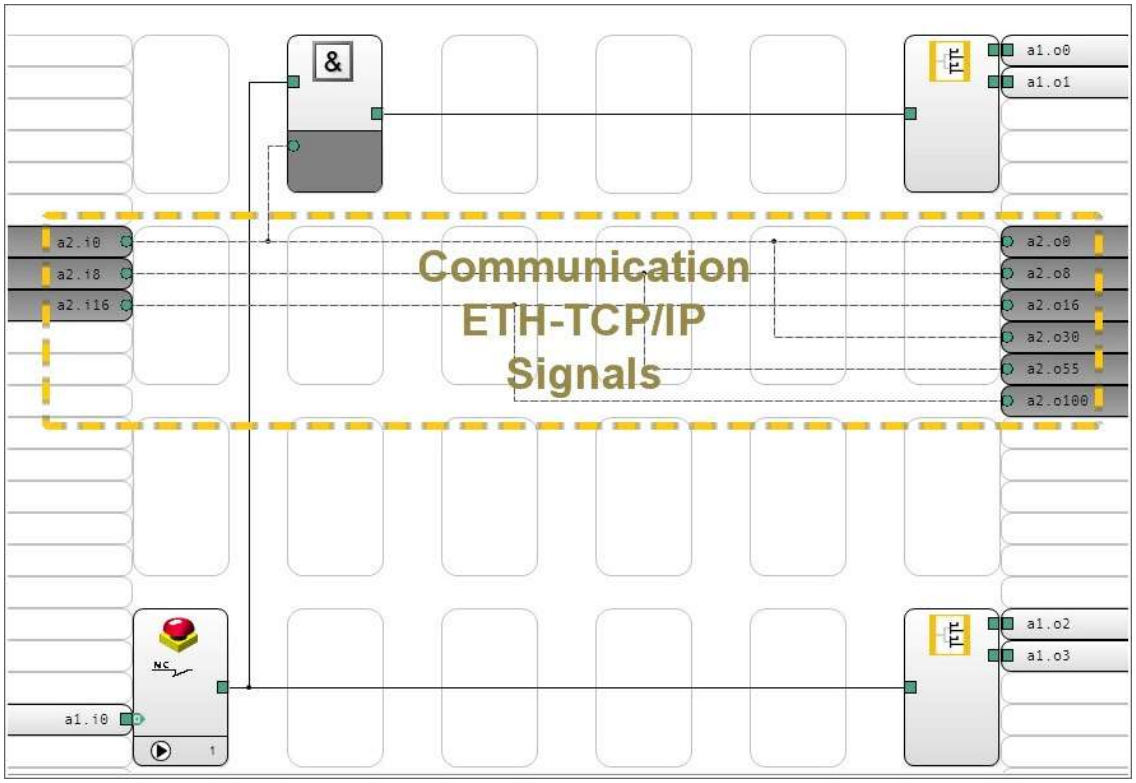


Fig. 36: PNOZmulti Configurator – Main Program “PNOZmulti_S7-PLC_ETH-TCP-IP_v2”

- Check the entered IP address (in example: 169.254.060.001), click at button “Online” and activate function “Dynamic Program Display”:



Fig. 37: PNOZmulti Configurator – Go “Online” and start “Dynamic Program Display”

If the lines of signals become green, than the Ethernet TCP/IP connection is online and the communication works.

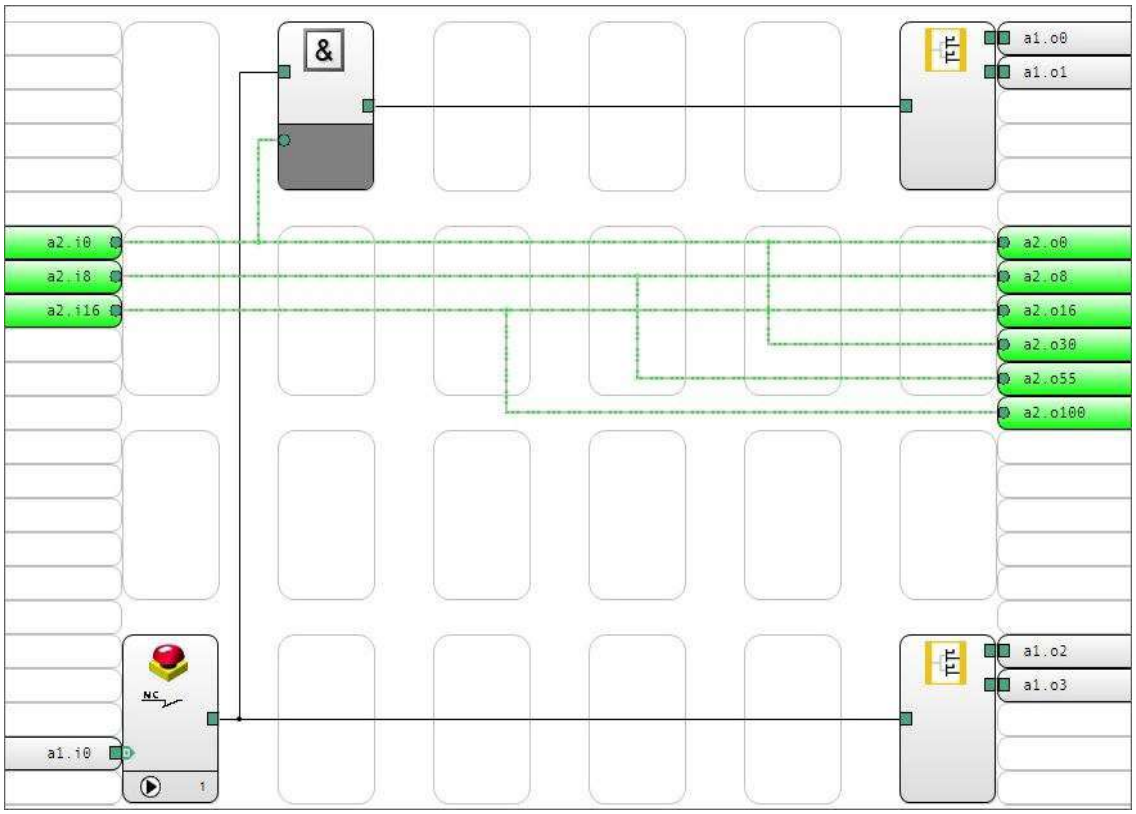


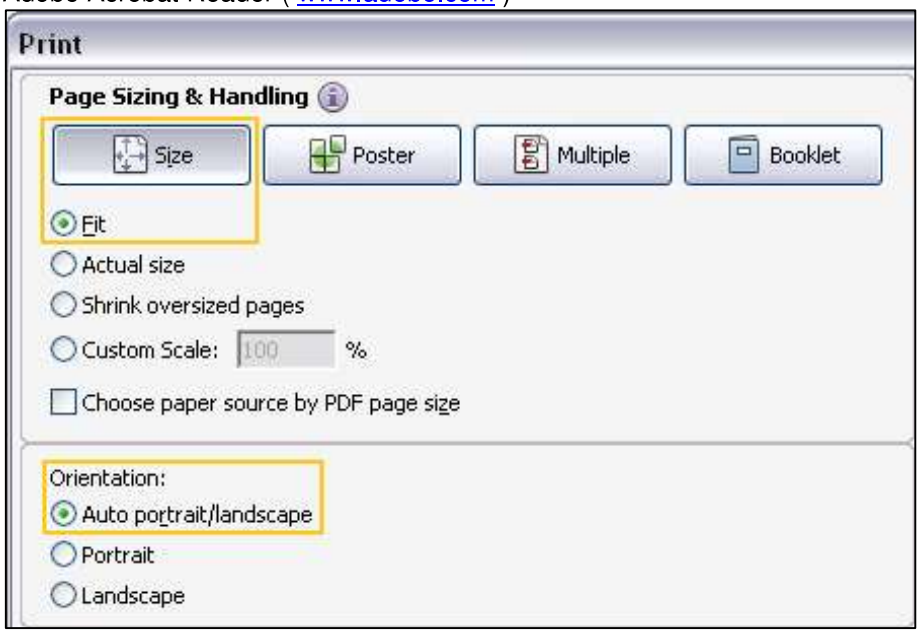
Fig. 38: PNOZmulti Configurator – Ethernet connection online and communication works

5. Table of figures

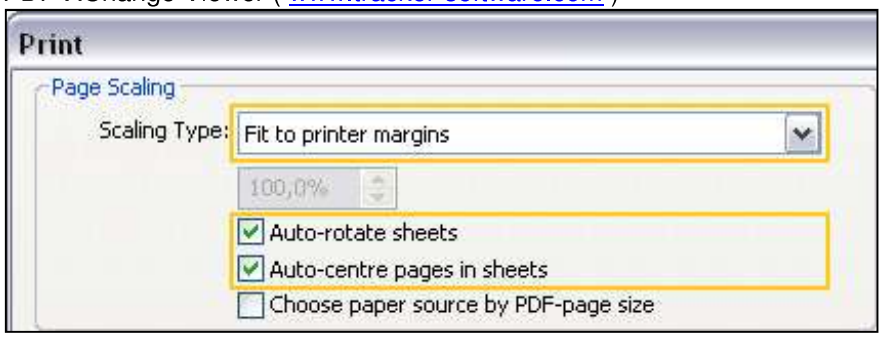
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Recommended printer settings

Adobe Acrobat Reader (www.adobe.com)



PDF-XChange Viewer (www.tracker-software.com)



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